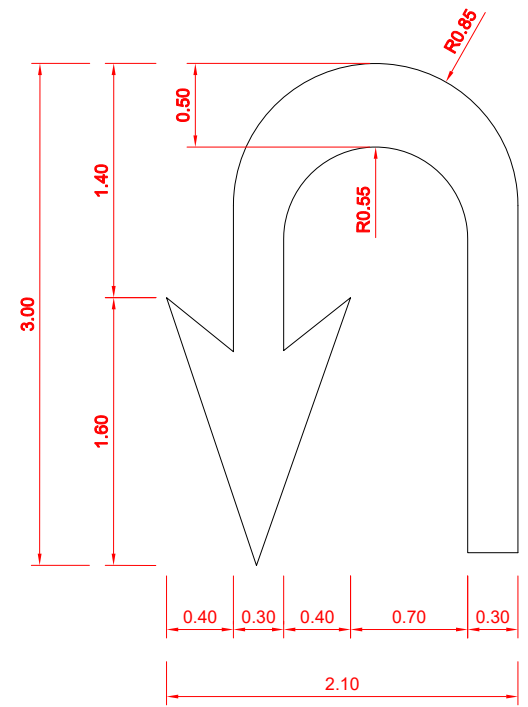
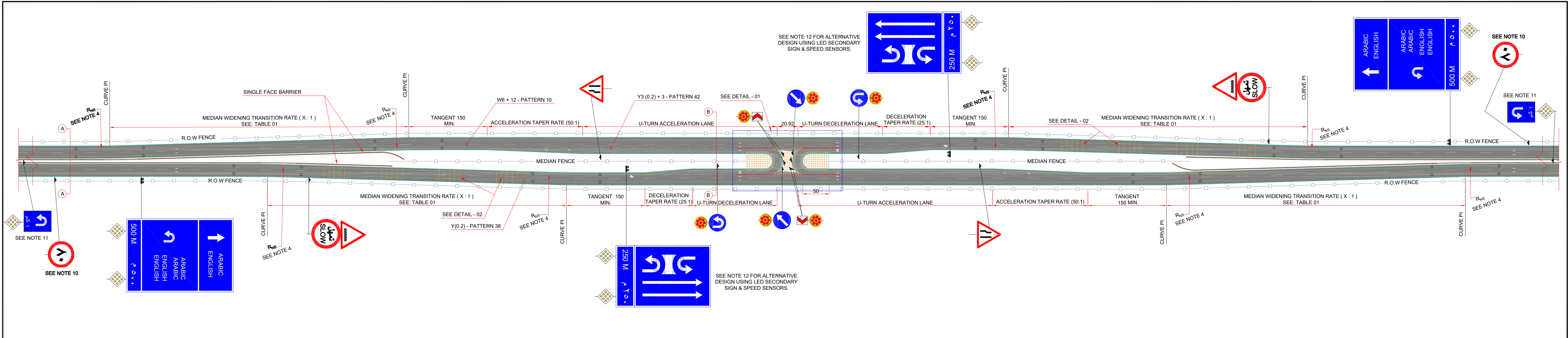
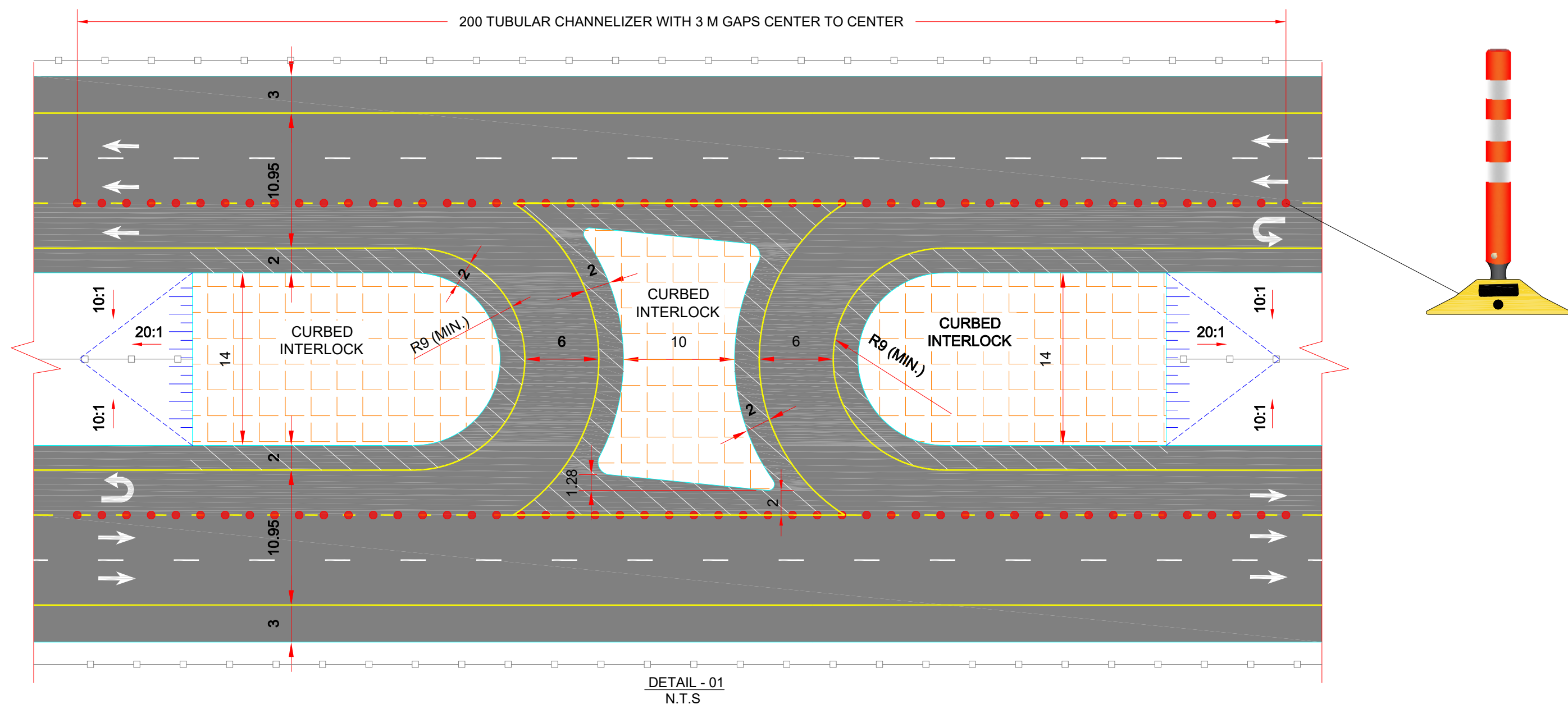




DETAIL - 03: U-TURN MARKING ARROW DETAIL
N.T.S



DETAIL - 01
N.T.S

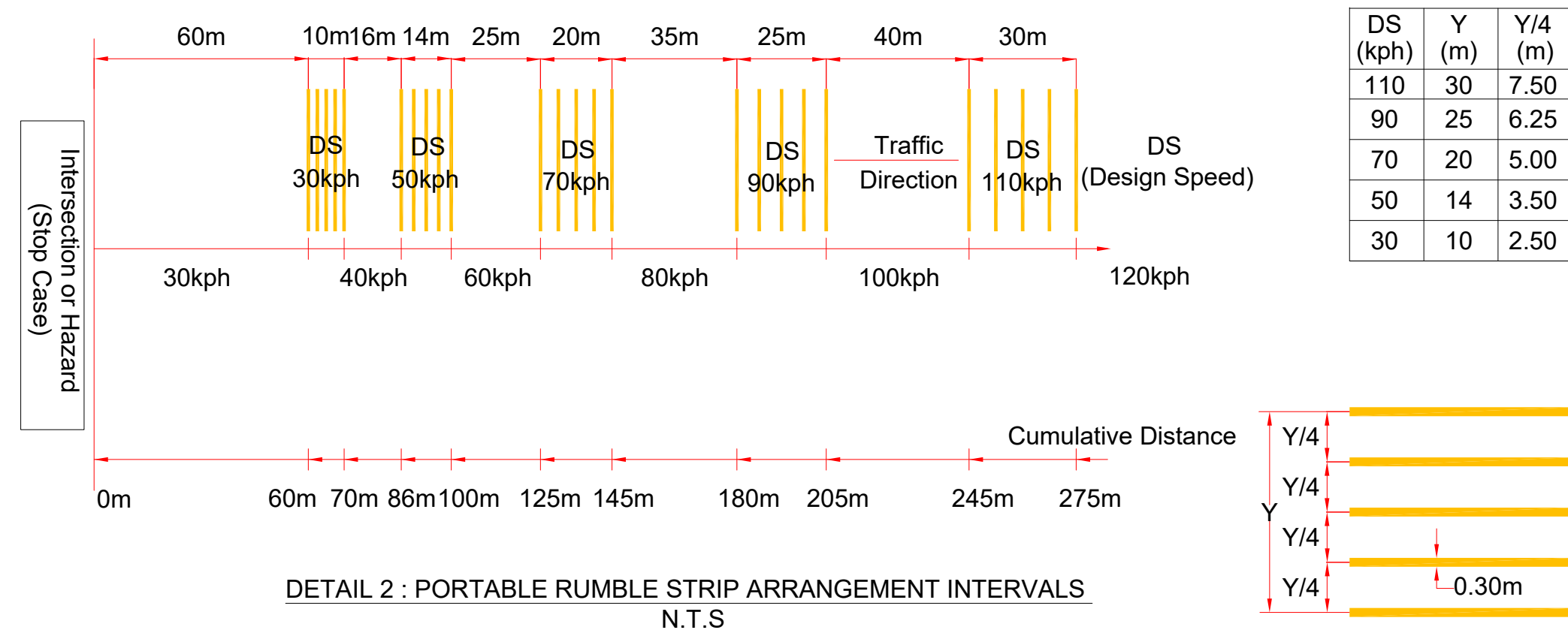
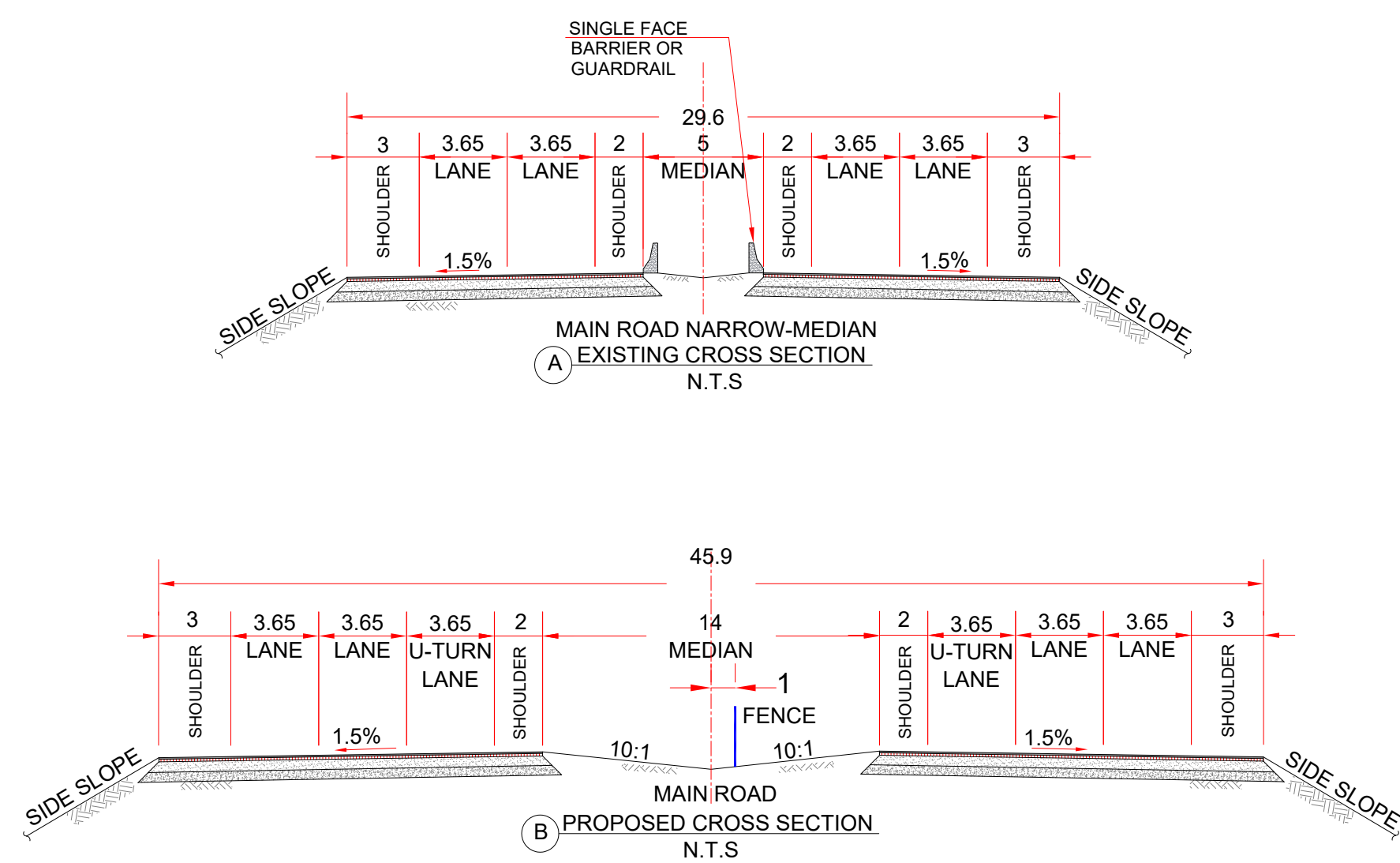
TABLE 01: MEDIAN WIDENING TRANSITION TAPER RATES.

DESIGN SPEED (KPH)	TRANSITION TAPER RATE (M:M)
60	25:1
70	45:1
80	50:1

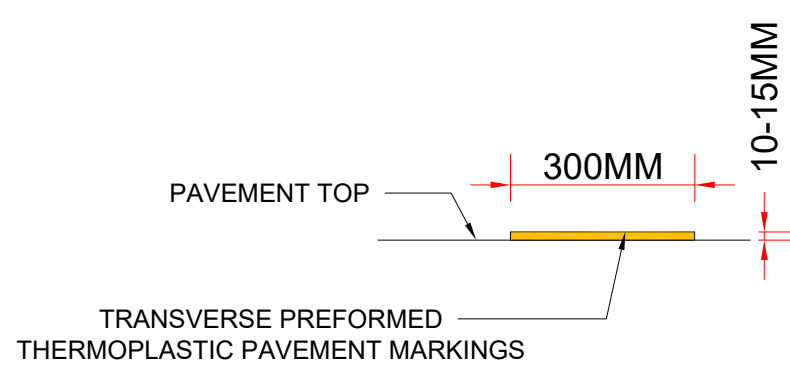
*THE SHOWN TAPER RATES DEFINE THE LONGITUDINAL DISTANCE (M) TRAVELED BY TRAFFIC VEHICLES TO MOVE LATERALLY BY (1 M) SHIFT DISTANCE OF THE TAPER .
* SHOWN TAPER RATES ARE NOT VALID FOR EITHER ACCELERATION NOR DECELERATION LANE TAPERS.
* OPERATION SPEED REDUCTION SHOULD BE APPLIED THROUGH MEDIAN TRANSITION SECTION .

NOTES :

- DIMENSIONS SHOWN ARE AS SPECIFIED AND NOT TO SCALE.
- SPEED-CHANGE LANE LENGTHS WILL VARY ACCORDING TO VARIANCE OF MAIN ROAD DESIGN SPEED, REFER TO DOCUMENT NUMBER: RD1-MOT-RSD-PDF-1812-001
- U-TURN AREA MUST BE FURNISHED WITH STREET LIGHTING, (IF ELECTRIC POWER LINES ARE NOT AVAILABLE ONSITE THEN, SOLAR POWERED LIGHTING SHOULD BE PROVIDED).
- TRANSITION CURVES RADII PREFERRED TO BE DESIGNED SO THAT NO SUPERELEVATION IS REQUIRED AS PER DESIGN SPEED REQUIREMENTS.
- U-TURN OPENINGS ARE TO BE AVOIDED ON CURVED SEGMENTS OF HIGHWAY ALIGNMENT EXCEPT FOR CURVES WITH LARGE RADII WHERE NO SUPERELEVATION IS REQUIRED ACCORDING TO MOT HDM DESIGN STANDARDS.
- NARROW MEDIAN (LESS THAN 9 M WIDTH) SHOULD BE ALWAYS PROTECTED WITH THE APPROPRIATE BARRIER SYSTEM.
- INTERSECTIONS AND DIRECT ACCESS ENTRANCES OR EXITS MUST BE AVOIDED THROUGH U-TURN AND TRANSITION ZONES.
- U-TURN ON HIGHWAYS IS NOT PREFERABLE AND SHOULD BE AVOIDED AS POSSIBLE. HOWEVER, AND IN CRITICAL NEED FOR U-TURN, CONTRACTOR MUST STUDY THE SITE CONDITIONS AND REQUIREMENTS PROVIDING MAXIMUM SAFETY LEVELS AND TOLERANT DESIGNS.
- OPERATION SPEEDS SHOULD BE REDUCED TO 80 KPH TO CONSTRUCT U-TURN IN A SPECIFIC LOCATION.
- EARLY TRAFFIC CALMING AND ENFORCEMENT MEASURES MUST BE INSTALLED / APPLIED TO REDUCE OPERATION SPEED TO 80 KPH AS MAXIMUM BEFORE SHOWN SIGN LOCATION.
- LARGE GUIDANCE SIGN COULD BE USED AS ALTERNATIVE ACCORDING TO NUMBER OF DESTINATION NAMES NEEDED TO BE SHOWN FOR U-TURN DIRECTION.
- LED SECONDARY SIGN IS AN EFFECTIVE ALTERNATIVE DISPLAYING "SLOW" MESSAGE (IN ARABIC AND ENGLISH) FOR SPEED VIOLATORS, PROVIDING THE MAIN SIGN WITH NECESSARY CALIBRATED SPEED SENSORS.



DETAIL 2 : PORTABLE RUMBLE STRIP ARRANGEMENT INTERVALS
N.T.S



KINGDOM OF SAUDI ARABIA
MINISTRY OF TRANSPORT



TITLE :
المخطط النموذجي لفتحة دوران للخلف لطريق سريع ذو جزيرة وسطية ضيقة
TYPICAL U-TURN DETAIL FOR NARROW-MEDIAN HIGHWAYS

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